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Stoia et al.

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(54) **INFOTAINMENT SYSTEM WITH AIR-VENT CONTROL**

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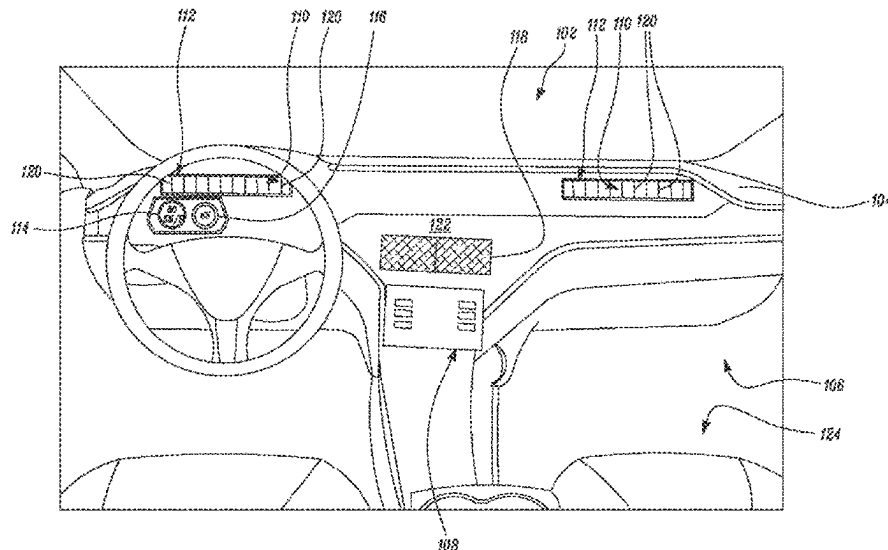
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(57) **ABSTRACT**

An air vent assembly includes a cover member having an inlet for allowing a flow of air to enter the cover member. The cover member also has a vent extending from the inlet for allowing a flow of air to exit the cover member. The vent includes a first vent portion, and a second vent portion located adjacent to the first vent portion. The first vent portion has a first set of flow guiding vanes disposed therein. The second vent portion has a second set of flow guiding vanes disposed therein. Each of the vanes is spaced apart from one another and pivotally coupled to a pair of opposing walls of the cover member. The first and second sets of flow guiding vanes are independently moveable with respect to each other using an actuating mechanism for providing a variety of flow-guiding positions one of which includes a split flow-guiding position.

23 Claims, 15 Drawing Sheets



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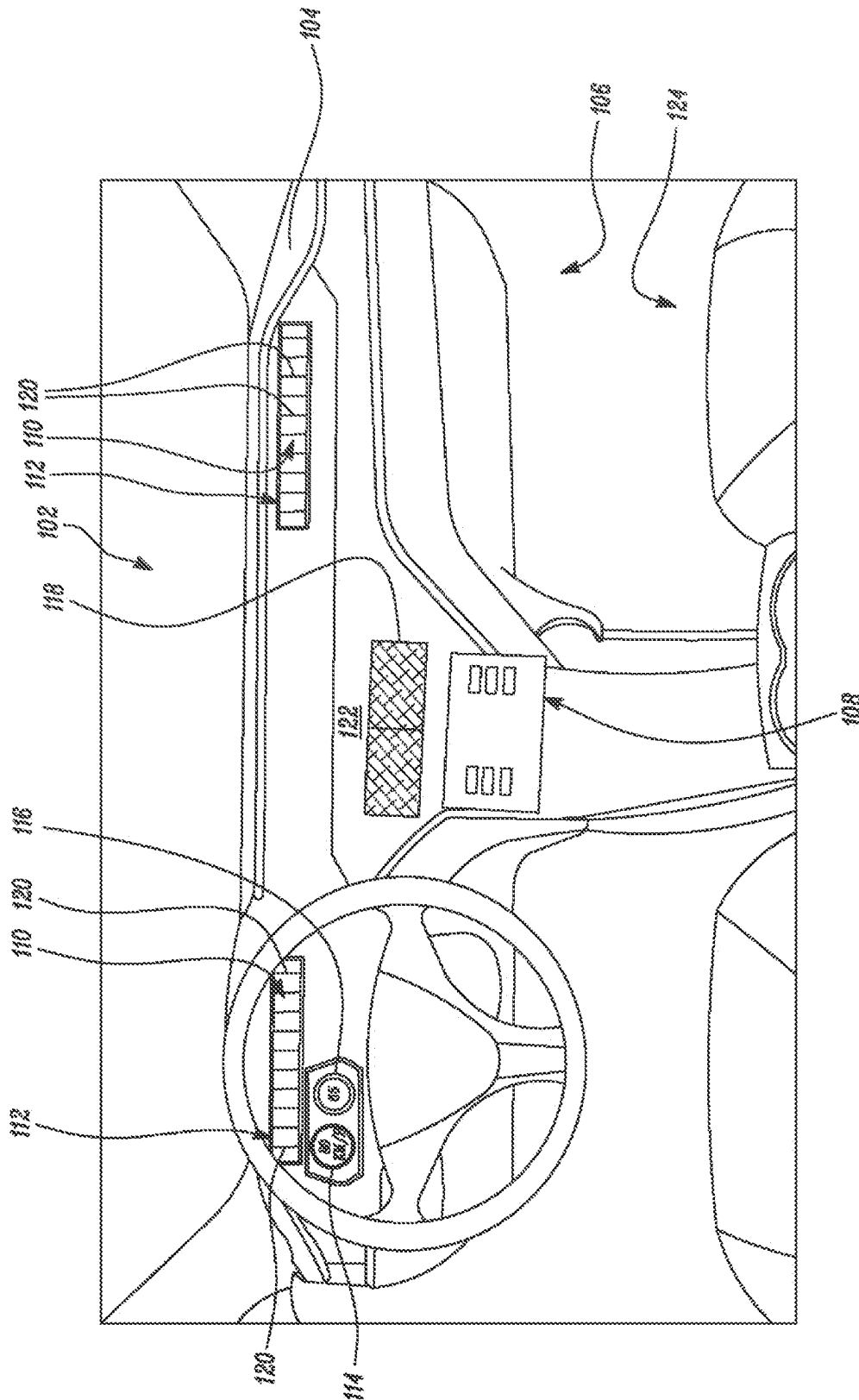


FIG. 1

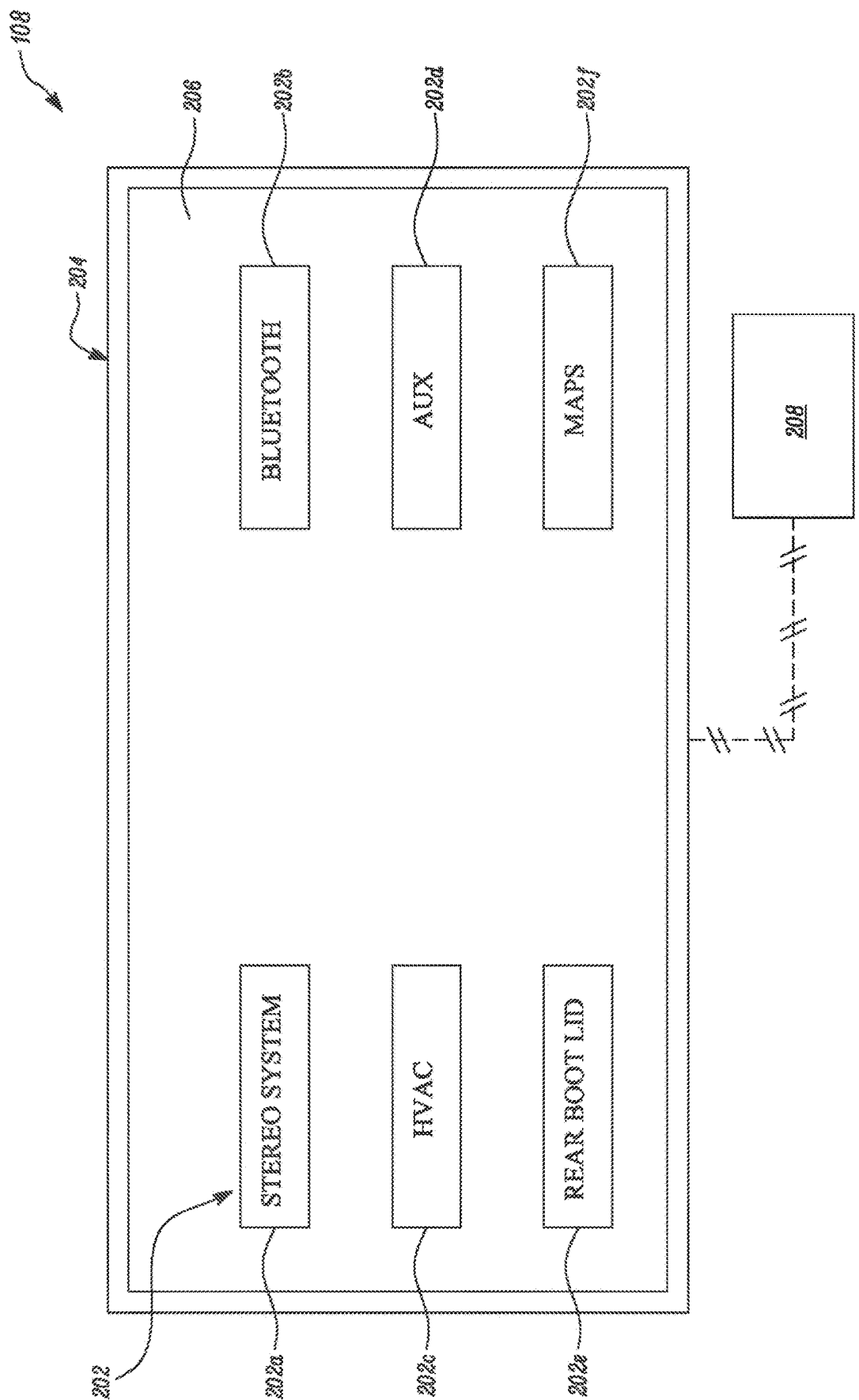


FIG. 2

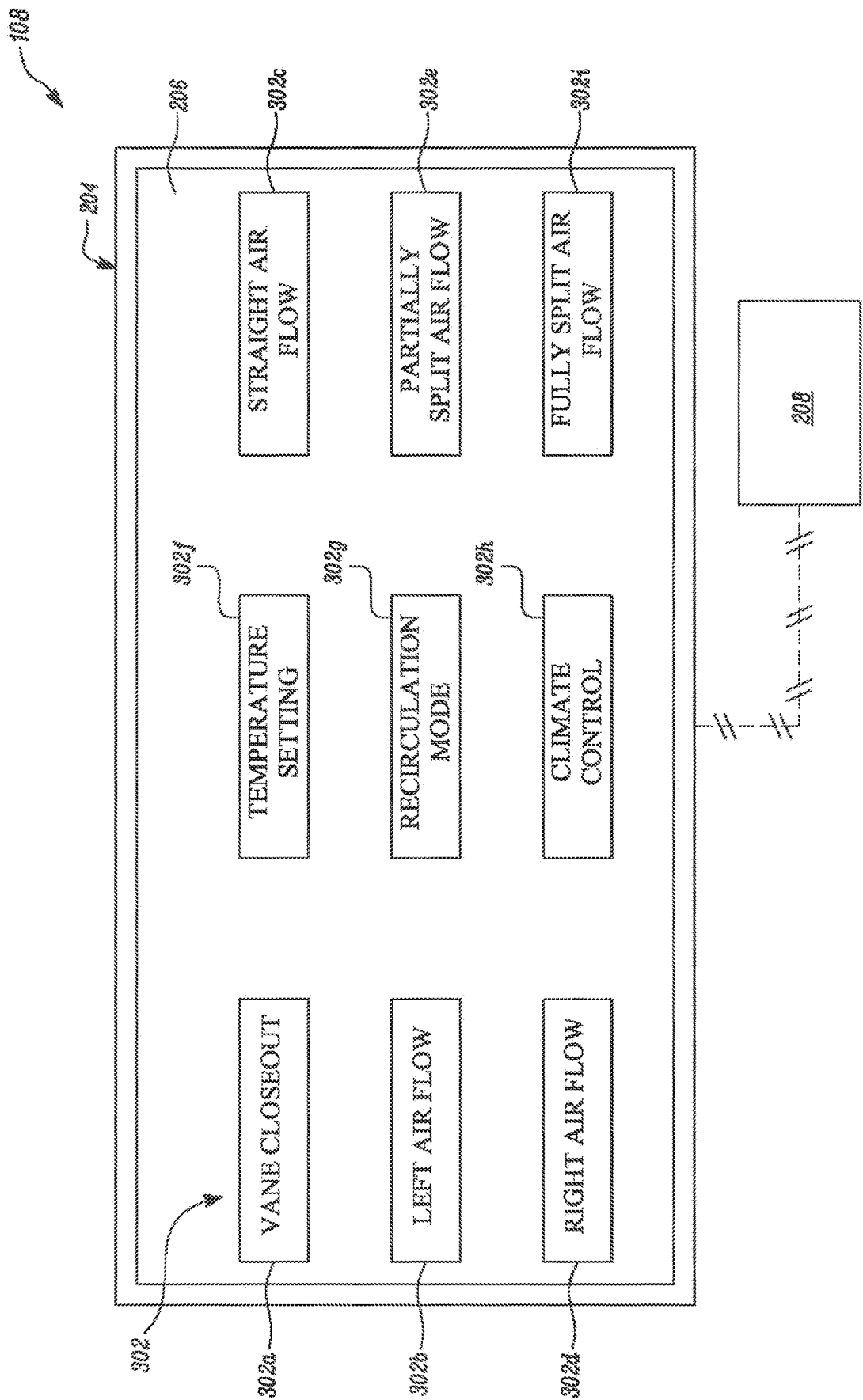


FIG. 3

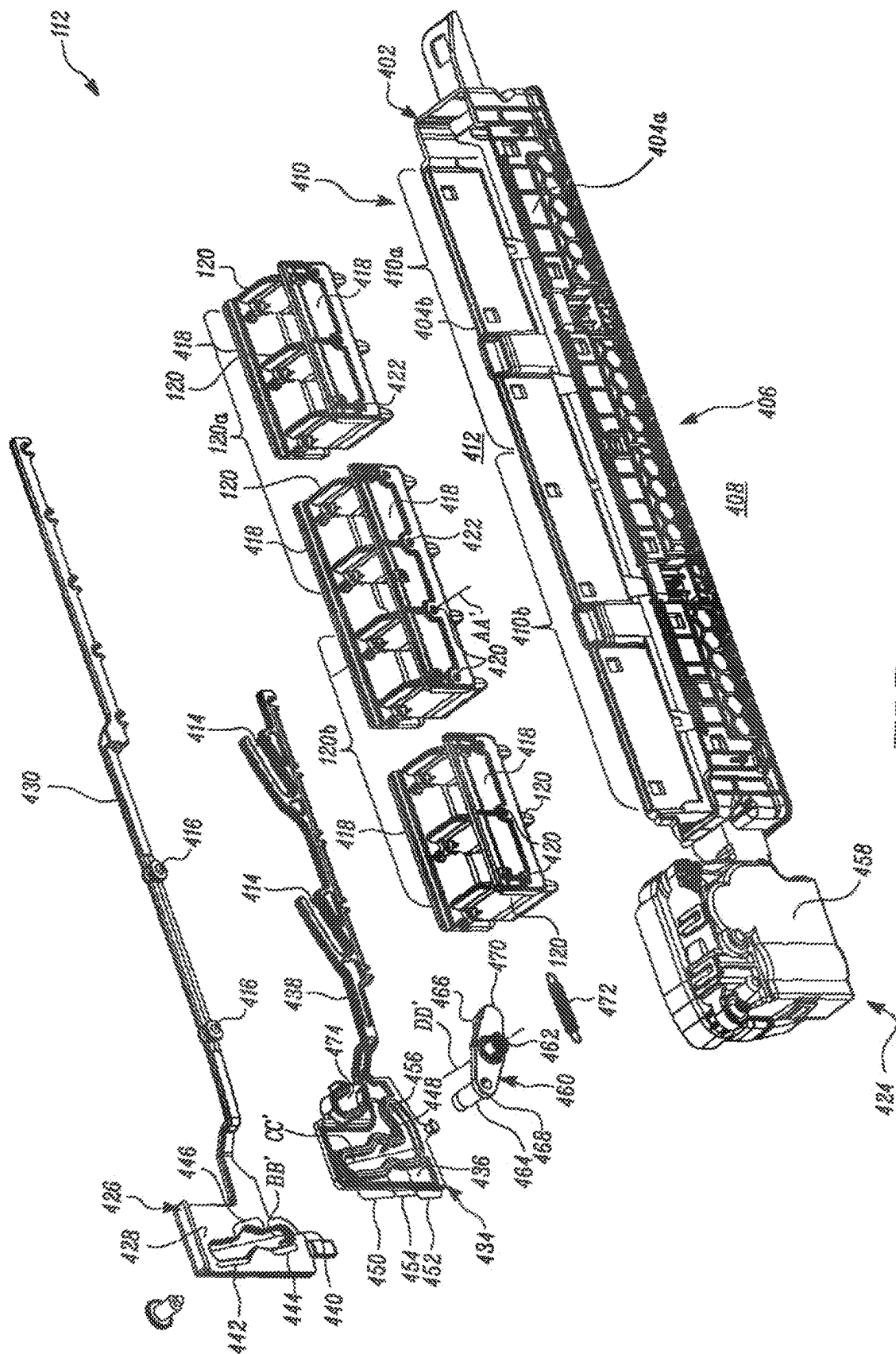


FIG. 4

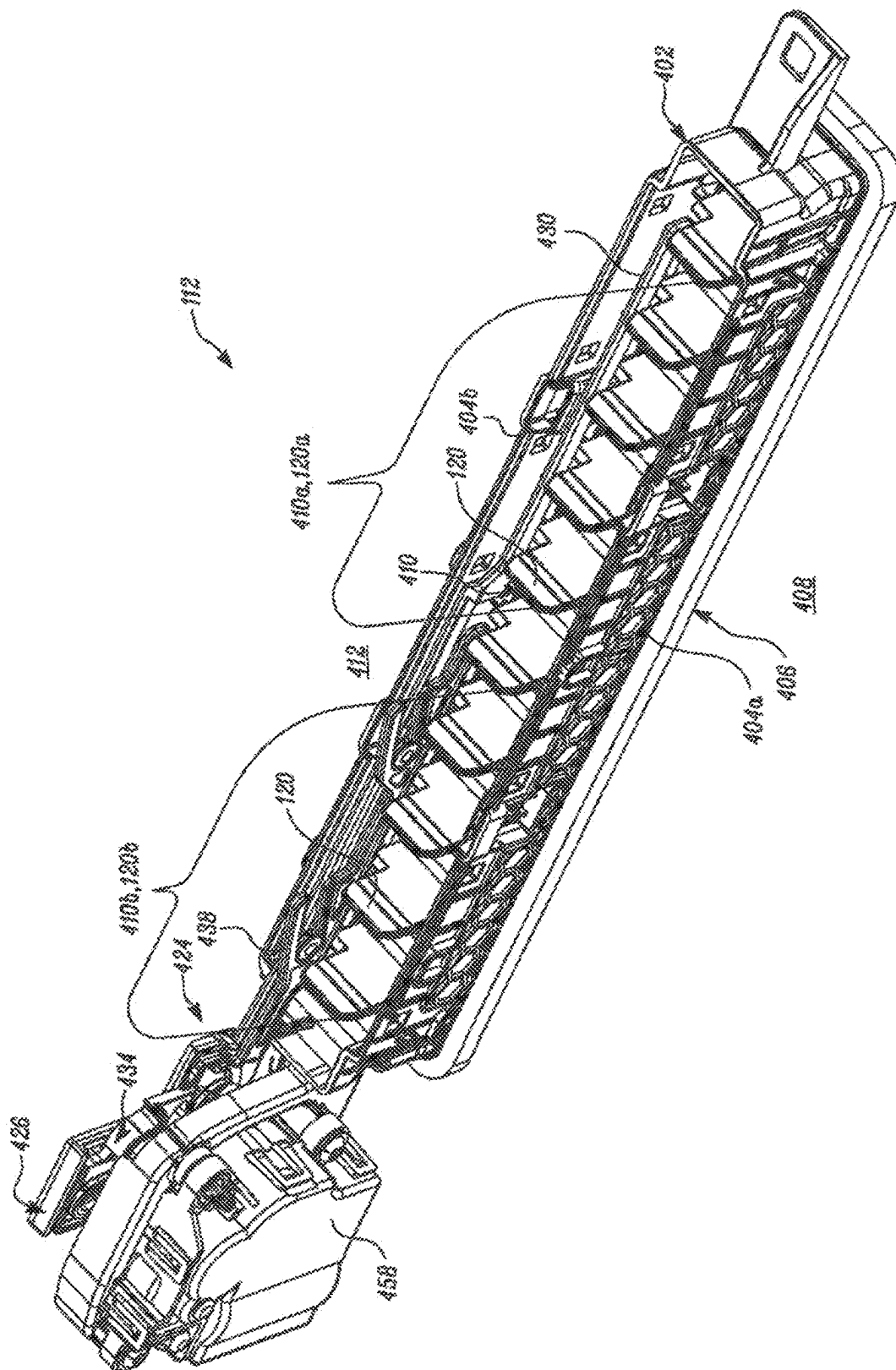
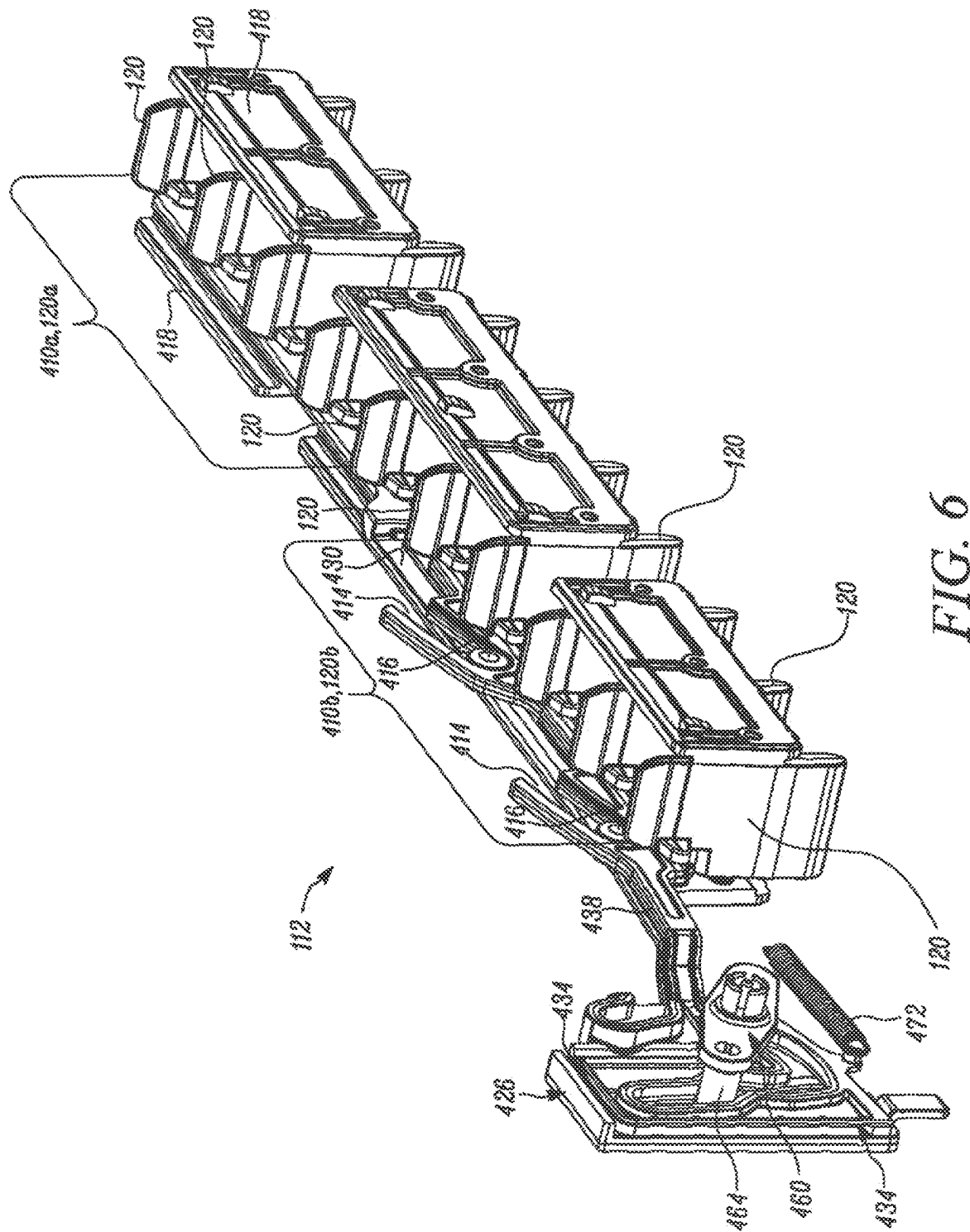


FIG. 5



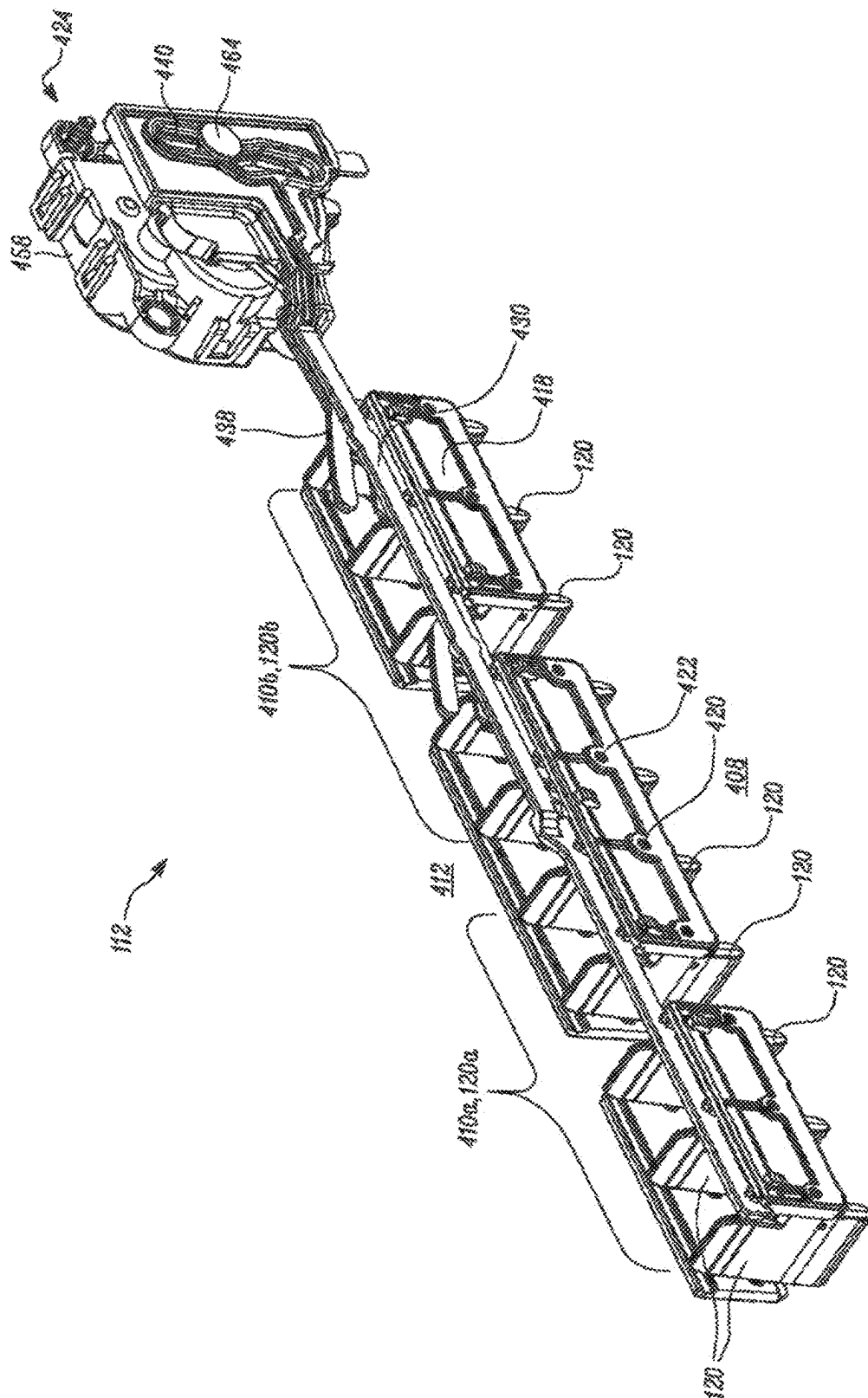


FIG. 7

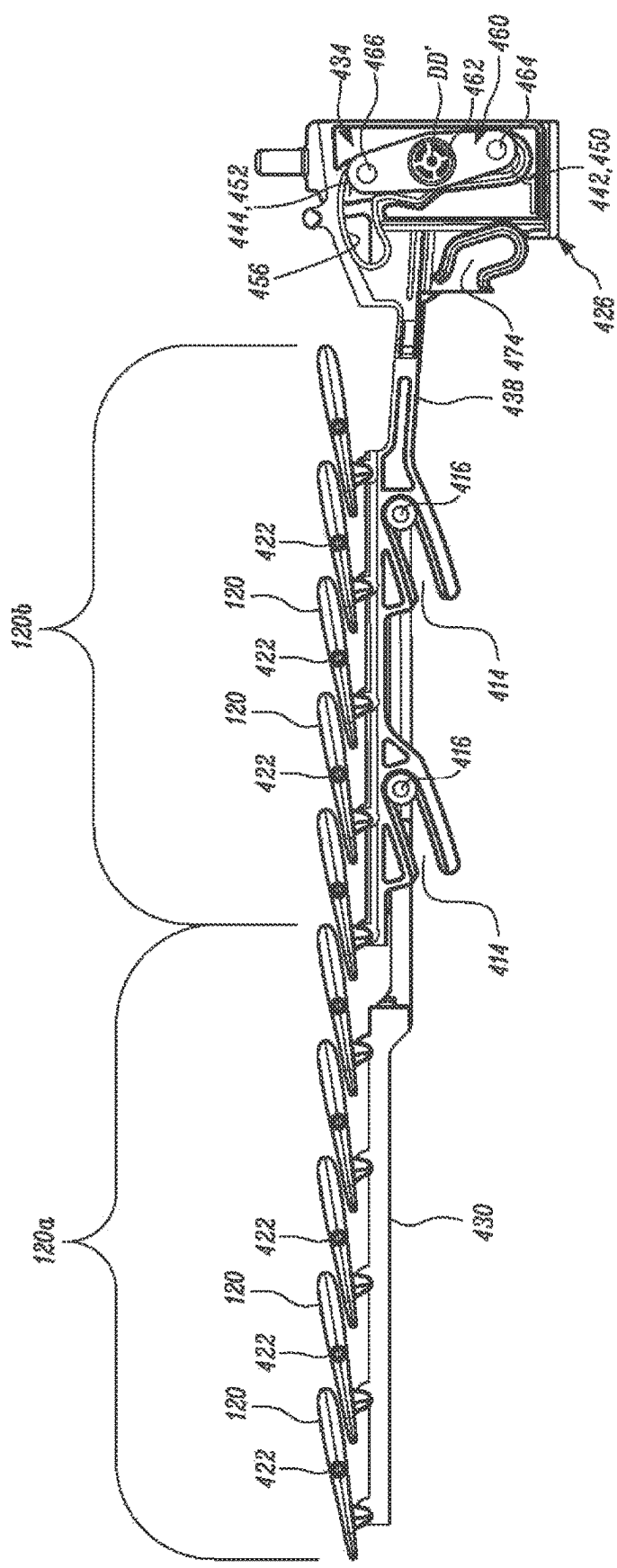


FIG. 8

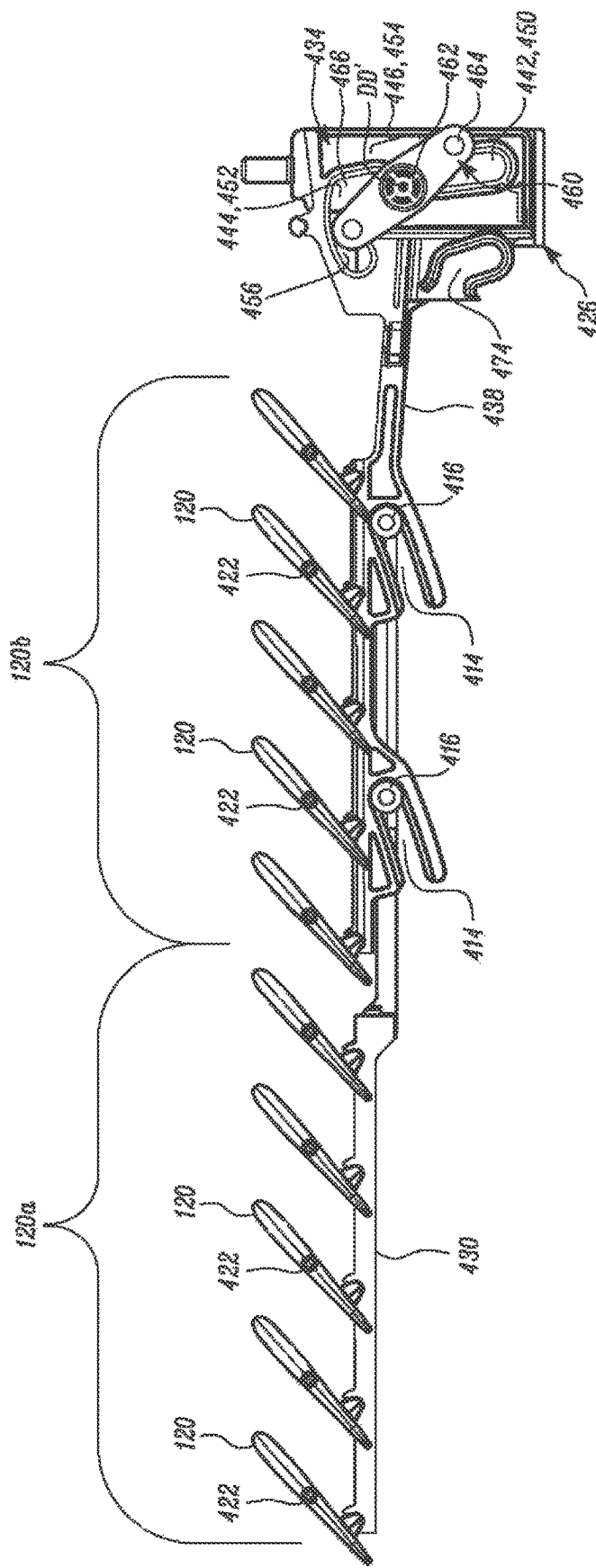


FIG. 9

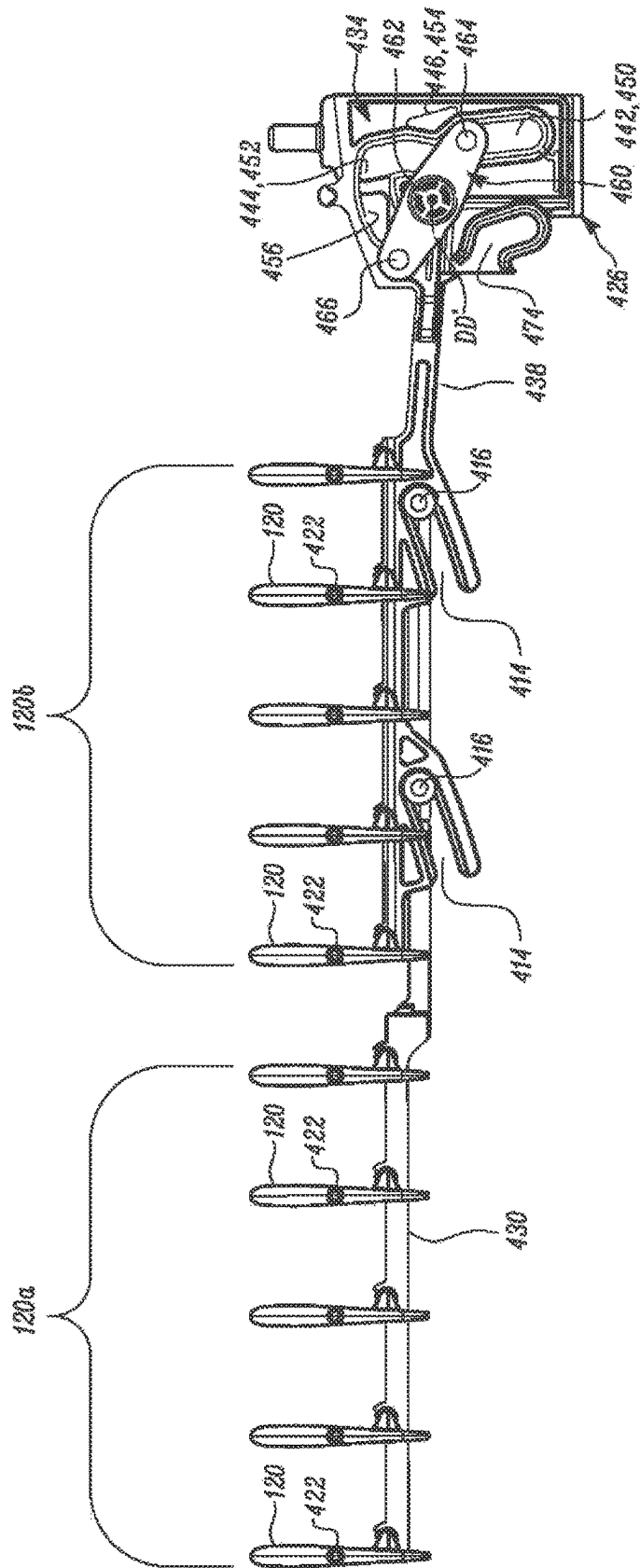
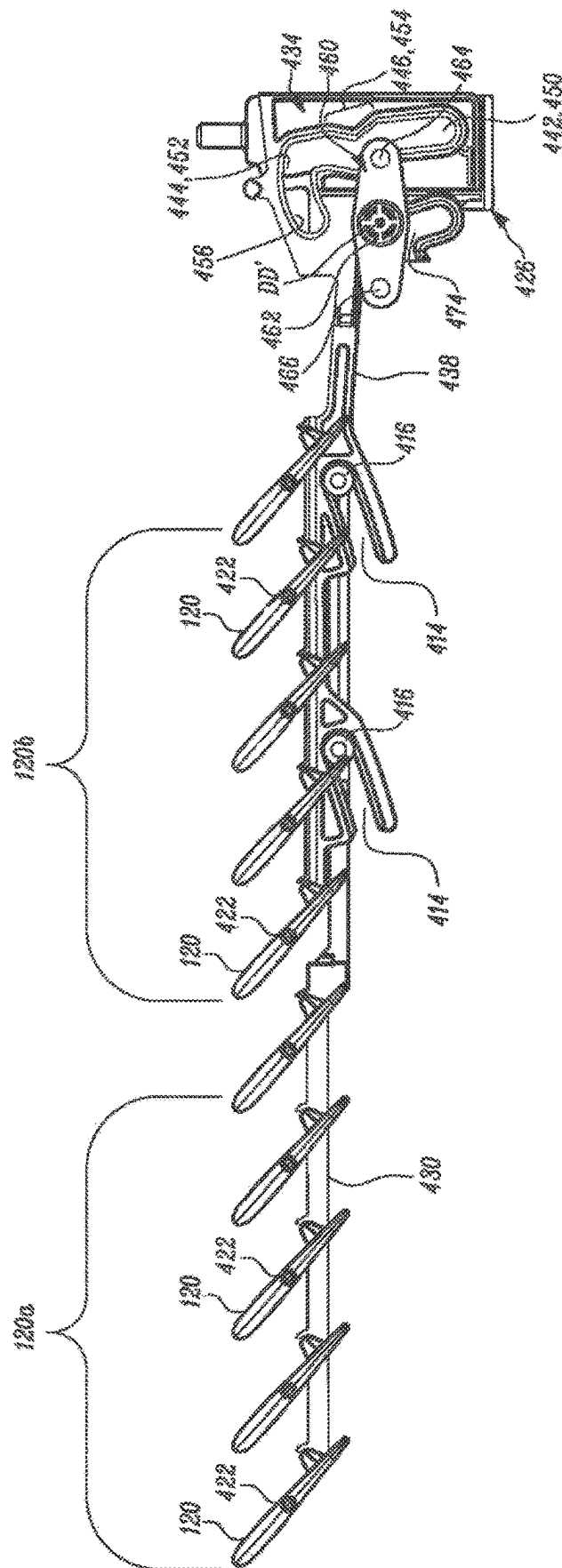


FIG. 10



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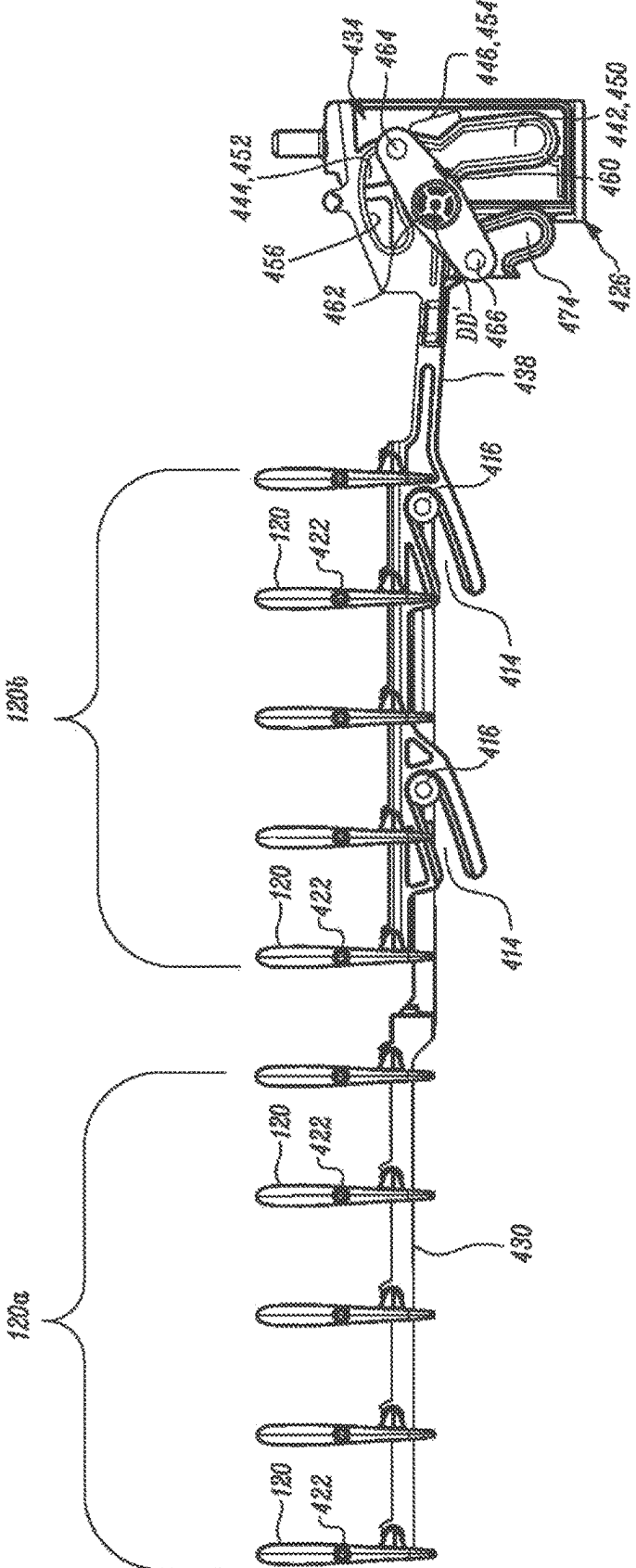


FIG. 12

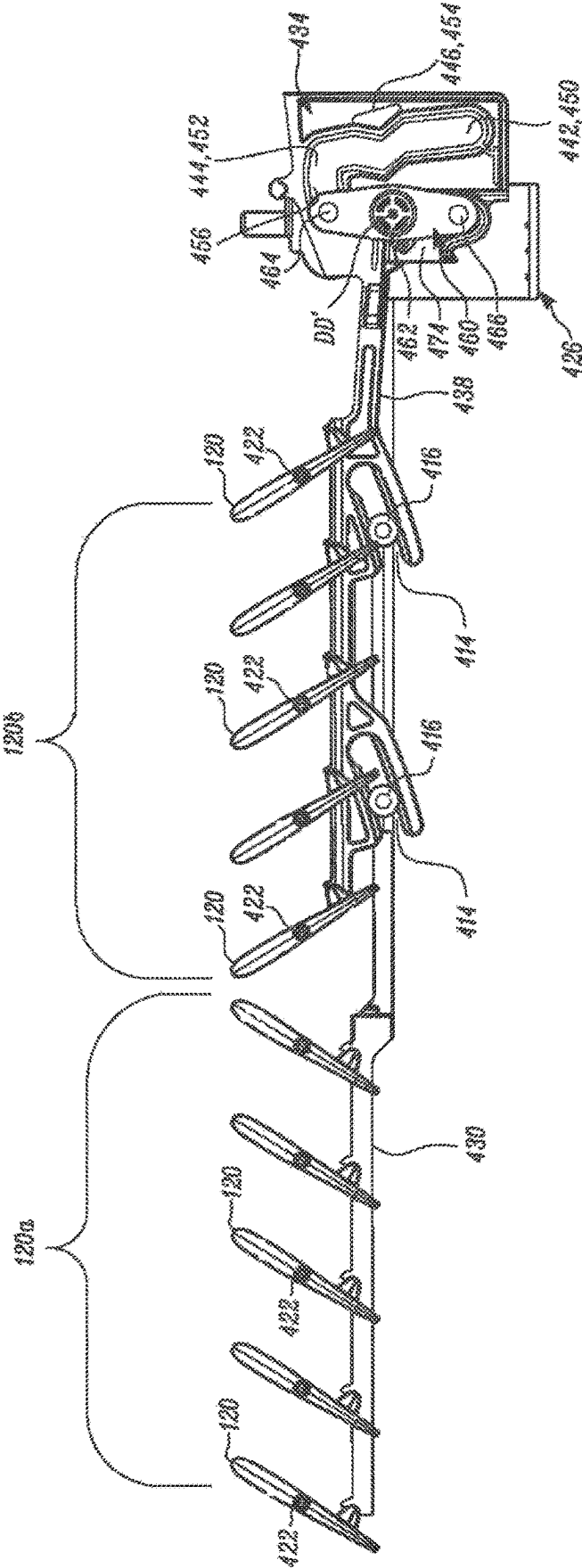


FIG. 13

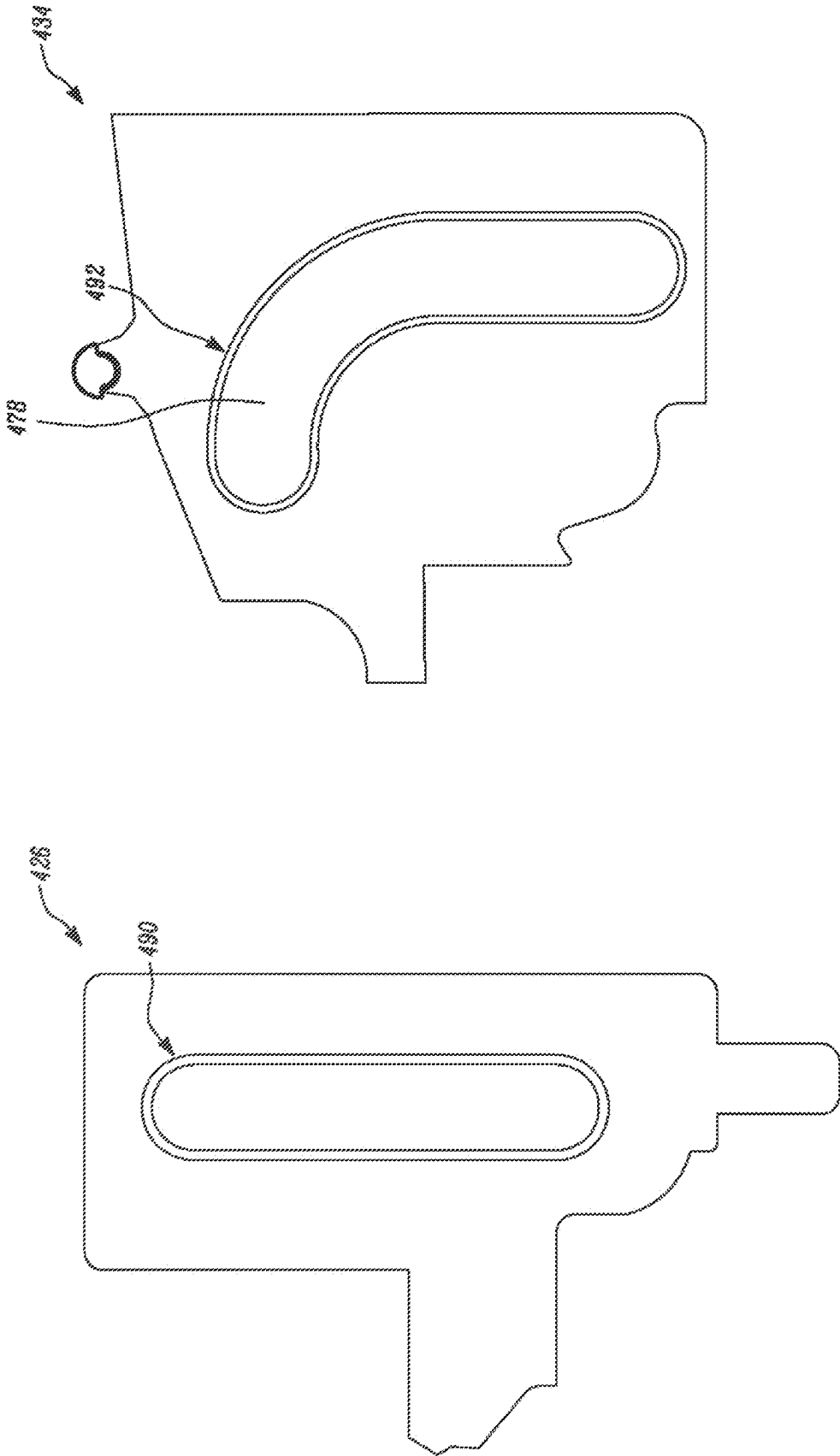


FIG. 14A

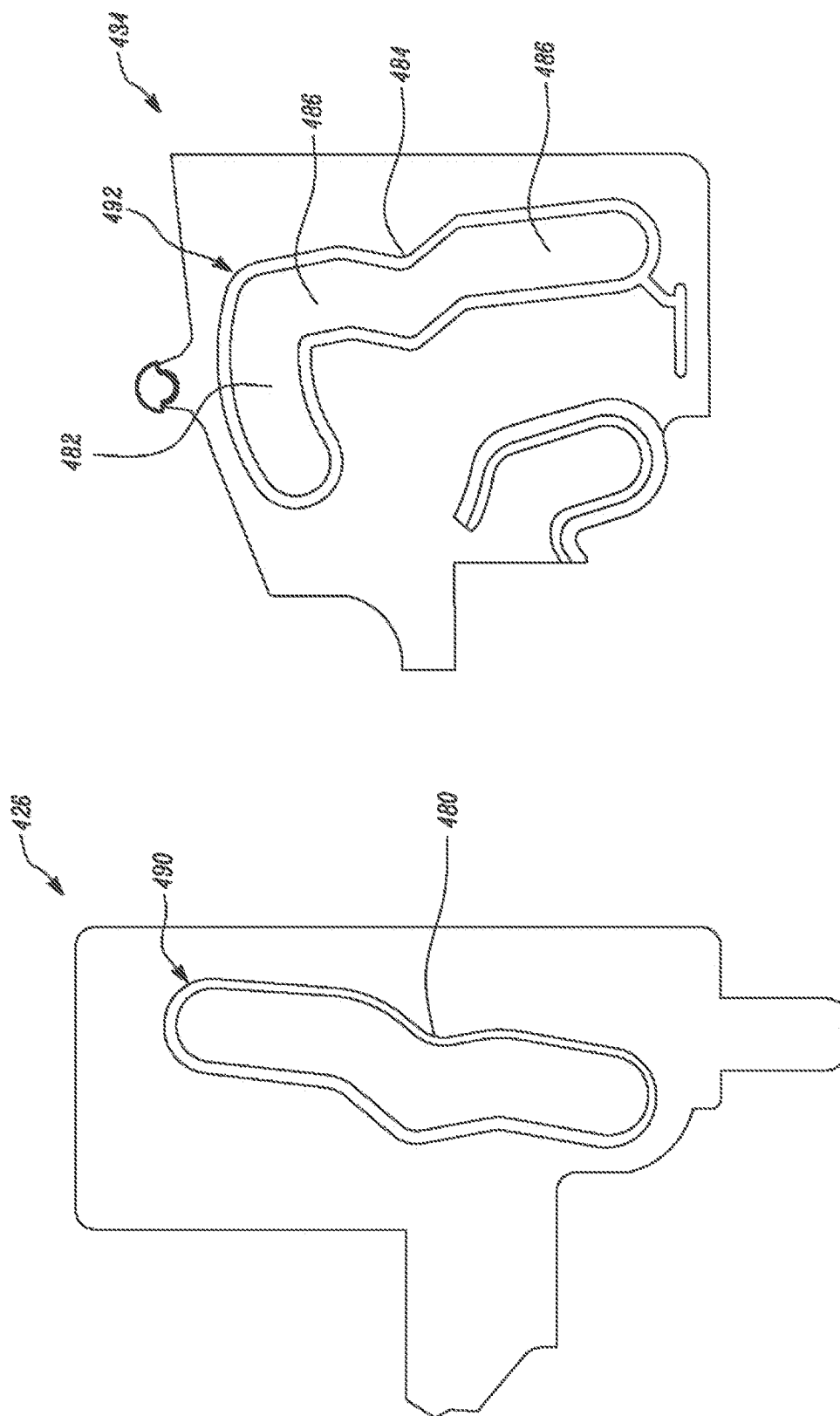


FIG. 14B

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INFOTAINMENT SYSTEM WITH AIR-VENT CONTROL

This application is a divisional application of U.S. application Ser. No. 15/374,035, entitled “Infotainment System with Air-Vent Control”, filed on Dec. 9, 2016, and issued as U.S. Pat. No. 10,752,088, each of which are hereby incorporated herein by reference in their entirety.

TECHNICAL FIELD

The present disclosure relates to an infotainment system that controls operation of an HVAC system in order to control air flow within a cabin space of a vehicle. More particularly, the present disclosure relates to an air vent assembly of an HVAC system that can be electronically controlled using an infotainment system.

BACKGROUND

It is well known in the art to provide an air vent assembly for delivering air from an HVAC system to a cabin space of a vehicle. In many cases, conventional air vent assemblies lack choices of desired air flow to ensure driver and passenger comfort when seated within the cabin space of the vehicle. Also, in other cases, many traditionally known air vent assemblies are typically controlled with input devices including, but not limited to, physical buttons and/or knobs thus making operation of the air vent assembly dependent upon a proper working condition of such input devices. However, as air flow within the vehicle is an important aspect to creating a comfortable environment for the driver and passengers of the vehicle, ensuring the reliability of the HVAC system, and in particular, a reliability of the air vent assembly may be desired. In addition, traditionally, vehicles have multiple vents for each of the driver and/or passenger. However, dashboard space is often at a premium due to other considerations, such as a larger windshield or room for meters.

Hence, there is a need for an air vent assembly that overcomes one or more of the aforementioned drawbacks. Further, there is need for improved control of the air vent assembly through an infotainment system which allows for easy and intuitive control to facilitate the desired condition of the cabin space within the vehicle.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 illustrates an interior of a vehicle showing the air vent assembly of an HVAC system that is capable of being controlled by an infotainment system according to certain embodiments of the invention.

FIG. 2 illustrates an interface of the infotainment system showing a variety of high-level menu options according to certain embodiments of the invention.

FIG. 3 illustrates an interface of the infotainment system showing a variety of low-level menu options for controlling the HVAC system according to certain embodiments of the invention.

FIG. 4 illustrates an exploded top perspective view of an air vent assembly of the HVAC system according to certain embodiments of the invention.

FIG. 5 illustrates an assembled top perspective view of the air vent assembly according to certain embodiments of the invention.

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FIG. 6 illustrates a partially assembled top perspective view of the air vent assembly without a cover member according to certain embodiments of the invention.

FIG. 7 illustrates a side perspective view of the air vent assembly without the cover member according to certain embodiments of the invention.

FIG. 8 illustrates a top view of the air vent assembly showing vanes disposed in a vane closeout position according to certain embodiments of the invention.

FIG. 9 illustrates a bottom view of the air vent assembly showing vanes disposed in a rightward flow-guiding position according to certain embodiments of the invention.

FIG. 10 illustrates a bottom view of the air vent assembly showing vanes disposed in an onward flow-guiding position according to certain embodiments of the invention.

FIG. 11 illustrates a bottom view of the air vent assembly showing vanes disposed in a leftward flow-guiding position according to certain embodiments of the invention.

FIG. 12 illustrates a bottom view of the air vent assembly showing vanes disposed in an onward flow-guiding position according to certain embodiments of the invention.

FIG. 13 illustrates a bottom view of the air vent assembly showing vanes disposed in a split flow-guiding position according to certain embodiments of the invention.

FIG. 14A illustrates a cam-groove geometries for the first and second actuating members of the air vent assembly according to certain embodiments of the invention.

FIG. 14B illustrates a cam-groove geometries for the first and second actuating members pursuant to the air vent assembly of FIG. 4 according to certain embodiments of the invention.

Embodiments of the present disclosure and their advantages are best understood by referring to the detailed description that follows. It should be appreciated that like reference numerals are used to identify like elements illustrated in one or more of the figures, wherein showings therein are for purposes of illustrating embodiments of the present disclosure and not for purposes of limiting the same.

DETAILED DESCRIPTION

The present disclosure relates to air vent assemblies. More particularly, the present disclosure relates to a particular air vent assembly and an infotainment system that controls an HVAC system, including the air vent assembly, in order to control air flow within a cabin space of a vehicle.

The air vent assembly according to the present disclosure introduces a new way of delivering airflow to a user. According to certain embodiments, the air vent assembly includes a first vent portion placed in front of the driver and a second vent portion that is placed in front of the passenger. These vent portions may be elongated as compared to traditionally known designs of air vent assemblies in that they have longer, more linear vents instead of more circular or boxy vents. Also, each of the vent portions from the air vent assembly of the present disclosure typically includes more than one vent for each of the driver and passenger.

Embodiments of the present disclosure enhance the capabilities of this air vent assembly to provide air flow in a leftward, rightward, onward, and a split configuration using a single rotary actuator. In certain embodiments, the user interface may also allow a user to define the degree to which the vents may cause air to flow in the rightward, leftward, or split direction. In certain other embodiments, the infotainment display may be configured to provide pre-defined degrees of movement to the vents for causing air to flow around the driver or passenger.

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According to certain embodiments, the driver and/or the passenger may control the vents using menu options provided on the user interface of the infotainment device. Some of the menu options provided on the user interface when actuated may configure the rotary actuator to close the vents, or cause the vents to accomplish a rightward air flow, a leftward air flow, an onward air flow, or a split air flow.

According to certain embodiments, the air vent assembly includes an actuating mechanism having a single rotary actuator therein for controlling a pair of actuating members—a first actuating member and a second actuating member. Each actuating member can control half of the vanes, although in certain other embodiments, each actuating member may control more or less than half of the vanes provided in the vent portions of the air vent assembly. Using the pair of actuating members together with the single rotary actuator, the vanes can be rotated about their respective rotational axes to close, or cause air to flow in the leftward, rightward, onward, or a split flow configuration in which flow of air is directed around the driver and/or the passenger of the vehicle. In certain embodiments, the user customizes the airflow, for example, the specific direction of the airflow. This can allow for a more comfortable experience for the occupants of a vehicle. For example, certain occupants may desire for air to be directed towards themselves while other occupants may desire the temperature of the vehicle cabin to be set to a certain value without having air directly impinging on themselves.

According to certain embodiments, the actuating mechanism includes a cam member that is rotatively driven by the rotary actuator in a range of 180 degrees, which in other embodiments could include other ranges of angular motion for accomplishing a movement of the vanes in their respective vane-closeout position, leftward flow guiding position, onward flow guiding position, rightward flow-guiding position, and a split flow-guiding position.

Reference will now be made in detail to specific aspects or features, examples of which are illustrated in the accompanying drawings. Wherever possible, corresponding or similar reference numbers will be used throughout the drawings to refer to the same or corresponding parts.

FIG. 1 illustrates an interior 102 of a vehicle 100 according to certain embodiments of the invention. As shown, a dash console 104 is disposed in an anterior region 106 of the vehicle 100. The dash console 104 is configured to provide a host of spaces for fitment of various devices including, but not limited to, an infotainment device 108, an air vent assembly 112 of an HVAC system 110, a speedometer 114, an odometer 116, a stereo system 118, and other like devices provided to the vehicle 100 depending on specific requirements of a vehicular application. Embodiments of the present disclosure are directed to the air vent assembly 112 of the HVAC system 110 and the infotainment device 108 that is configured to operatively control, amongst other things, the movement of vanes 120 in the air vent assembly 112 of the HVAC system 110.

FIG. 2 illustrates an interface 206 of the infotainment device 108 showing a variety of high-level menu options 202 according to certain embodiments of the invention. Each of the menu options 202 is provided on the user interface 206 of a display 204 associated with the infotainment device 108. This way, a user may select an appropriate menu option 202 on the user interface 206 of the infotainment device 108. Consequently, a processor 208 of the infotainment device 108 detects the menu option 202 that is being selected by the user and provides appropriate control signals to a desired device present in the vehicle 100 and

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corresponding with the selected menu option 202 so that an operation of the desired device is controlled. As disclosed earlier in this embodiment, the user input is provided by selecting from the menu options 202 listed on the user interface 206. According to certain other embodiments of this invention, the user can provide a voice command to the user interface 206 in lieu of physically selecting a menu option to control the air flow in vehicle 100.

As shown in the illustrated embodiment of FIG. 2, a host of high-level menu options 202 are shown provided on the user interface 206 of the infotainment device 108. Each of these high-level menu options 202 are user selectable for controlling an operation of the desired device. For example, as shown in FIG. 2, a menu option 202a when selected may be configured to direct the processor 208 to turn ON the stereo system 118. Likewise, a menu option 202b when selected may be configured to direct the processor 208 to turn ON a Bluetooth device while a menu option 202c when selected by the user would direct the processor 208 to open a rear boot lid of the vehicle 100. Similarly, a menu option 202d when selected may configure the processor 208 to turn ON the HVAC system 110 of the vehicle 100. Other menu options including, but not limited to, the menu options 202e and 202f may, additionally or optionally, be provided on the user interface 206 to facilitate AUX input at the stereo system 118 and enable navigation maps at the user interface 206 itself respectively.

It may be noted that the processor 208 may be a single microprocessor or multiple microprocessors that include components for performing functions consistent with the present disclosure. Numerous commercially available microprocessors can be configured to perform the functions of the processor 208 disclosed herein. It should be appreciated that the processor 208 could readily be embodied in a general purpose microprocessor capable of controlling numerous functions associated with each of the devices present in the vehicle 100. The processor 208 may also include a memory, a secondary storage device, and any other components for running an application. Various circuits may be associated with the processor 208 such as power supply circuitry, a solenoid driver circuitry, a signal conditioning circuitry for e.g., an analog-to-digital converter circuitry, and other types of circuitry. Various routines, algorithms, and/or programs can be programmed within the processor 208 for execution thereof. Moreover, it should be noted that the processor 208 disclosed herein may be a stand-alone processor 208 or may be configured to co-operate with existing processor/s, for example, an electronic control module (ECM) (not shown) provided to the vehicle 100 to perform functions that are consistent with the present disclosure.

Referring to FIG. 3, a host of low-level menu options 302 are displayed on the user interface 206 of the infotainment device 108 in accordance with certain embodiments of the invention. In particular, the low-level menu options 302 shown in FIG. 3 may be displayed on the user interface 206 in response to the user having previously selected the HVAC menu option 202d listed in the high-level menu options 202 from FIG. 2. As shown in FIG. 3, the low-level menu options 302 may include, but are not limited to, a temperature setting menu option 302f, a menu option 302g for enabling a recirculation mode of air conditioning in the vehicle 100, a menu option 302h for enabling climate control in the vehicle 100.

According to specific embodiments of this invention, at least some low-level menu options 302 are additionally provided to a user for particularly controlling the vanes 120

of the air vent assembly 112. As shown in FIG. 3, these low-level menu options 302 include a menu option 302a for directing the vanes 120 to move to a vane closeout position, a menu option 302b for directing the vanes 120 to move to a leftward flow-guiding position, a menu option 302c for directing the vanes 120 to move to an onward flow-guiding position, a menu option 302d for directing the vanes 120 to a rightward flow-guiding position, a menu option 302e for directing the vanes 120 of the air vent assembly 112 to a partially-split flow-guiding position, and another menu option 302i for directing the vanes 120 of the air vent assembly 112 to a fully-split flow-guiding position.

Referring back to FIG. 1, an air vent assembly 112 is shown typically disposed in front of the driver seat and front-passenger seat on the dash console 104. Therefore, references to the flow-guiding positions of the vanes 120 such as closeout, leftward, rightward, onward, and split are to be construed as being taken in relation to the seats or occupants of the vehicle 100 who would typically be located in a front portion 124 of the vehicle 100. Additional detail for each of the aforementioned flow-guiding positions appears below in the detailed description of the other figures.

Moreover, although it is disclosed in the illustrated embodiment of FIG. 1 that two air vent assemblies are disposed on the dash console 104—one in front of the driver seat and one in front of the passenger seat—it may be noted that the number and position of the air vent assembly 112 or assemblies within the interior 102 of the vehicle 100 is non-limiting of this invention. Other alternative positions within the interior 102 of the vehicle 100 may include, but is not limited to, the center of the console, a right inner side of the vehicle 100, a left inner side of the vehicle 100, a top inner side of the vehicle 100, or stretching from the right side of the vehicle 100 to the left side of the vehicle 100. Such alternative positions may be contemplated depending on specific requirements of a vehicular application.

FIG. 4 shows an exploded top perspective view of the air vent assembly 112 in accordance with certain embodiments of this invention. As shown, the air vent assembly 112 includes a cover member 402. An outer shape and size of the cover member 402 may be selected so as to correspond with a size and shape of a space provided by the dash console 104 of the vehicle 100 (refer to FIG. 1). In instances where the air vent assembly 112 is to be installed in a location other than the space provided by the dash console 104, for example, the left inner side, the right inner side, or the top inner side of the vehicle 100, the outer size and shape of the cover member 402 may be suitably selected to correspond with a size and shape of that space.

With continued reference to FIG. 4, the cover member 402 has a pair of elongated walls 404a and 404b disposed in an opposing relation to one another. The cover member 402 has an inlet 406 defined on an aft side 408 thereof. The inlet 406 allows a flow of air to enter the cover member 402, particularly, when the HVAC system 110 is turned ON in response to the user input at the interface being indicative of the HVAC menu option 202d.

The cover member 402 also has a vent 410 extending from the inlet 406 toward a fore side 412 of the cover member 402. The vent 410 is configured to allow a flow of air to exit the cover member 402. The vent 410 consists of a first vent portion 410a and a second vent portion 410b located adjacent to the first vent portion 410a. As shown, the first vent portion 410a has vanes 120 disposed therein. For purposes of simplicity, the vanes 120 from the first vent portion 410a are hereinafter referred to as ‘the first set of flow-guiding vanes’ and denoted with like alpha-numeral

‘120a’. Likewise, the second vent portion 410b has a second set of flow-guiding vanes disposed therein, which for purposes of simplicity and differentiation from the first set of flow-guiding vanes 120a is denoted with like alpha-numeral ‘120b’.

In the illustrated embodiment of FIG. 4, the first set of flow-guiding vanes 120a is shown to consist of five vanes 120. Likewise, the second set of flow-guiding vanes 120b consists of five vanes 120 therein. It may be noted that a number and distribution of vanes 120 between the first and second sets of flow-guiding vanes 120a, 120b is merely a matter of design choice and hence, the number and distribution of vanes 120 between the first and second sets of flow-guiding vanes 120a, 120b may vary from one application to another depending on specific requirements of an application. Moreover, although the number of vanes 120 in the air vent assembly 112 of the present disclosure is limited to five in each of the first and second sets of flow-guiding vanes 120a, 120b, it will be acknowledged by persons skilled in the art that an unequal number of vanes 120 may be present in the first and second sets of flow-guiding vanes 120a, 120b depending on specific requirements of an application.

Further, each of the vanes 120 from the first and second set of flow-guiding vanes 120b are spaced apart from one another. Moreover, the vanes 120 from the first and second vent portions 410a, 410b extend at least partially between the aft and fore sides 408, 412 of the cover member 402. Furthermore, each vane 120 from the first and second sets of flow-guiding vanes 120a, 120b are pivotally coupled to the pair of mutually opposing elongated walls 404a, 404b of the cover member 402. In the illustrated embodiment of FIGS. 4-7, three sets of cover plates 418 are provided to establish a pivotal connection of each vane 120 from the first and second sets of flow-guiding vanes 120a, 120b with the pair of mutually opposing elongated walls 404a, 404b of the cover member 402. As best shown in FIG. 4, each pair of mutually opposing cover plates 418 defines multiple corresponding recesses 420. Each of the recesses 420 on the cover plates 418 receives a pivot pin 422 associated with the vane from either the first or second set of flow-guiding vanes 120b.

According to certain embodiments of this invention, the first set of flow-guiding vanes 120a and the second set of flow-guiding vanes 120b are independently moveable with respect to each other using an actuating mechanism 424 that is coupled to each of the first and second sets of flow-guiding vanes 120a, 120b. The actuating mechanism 424 includes a first actuating member 426. The first actuating member 426 has a first base portion 428, and a first link member 430 extending from the first base portion 428. The first link member 430 is engaged with each vane 120 from the first set of flow-guiding vanes 120a. The first link member 430 is operable to pivot each vane 120 from the first set of flow-guiding vanes 120a about its corresponding rotational axis AA’.

Referring to FIGS. 4-7, the actuating mechanism 424 further includes a second actuating member 434 located adjacent to the first actuating member 426. The second actuating member 434 has a second base portion 436 that is disposed below the first base portion 428 of the first actuating member 426. The second actuating member 434 also has a second link member 438 extending from the second base portion 436. The second link member 438 is pivotally engaged with each vane 120 from the second set of flow-guiding vanes 120b. The second link member 438 is selec-

tively operable to pivot each vane **120** from the second set of flow-guiding vanes **120b** about its corresponding rotational axis AA'.

Additionally, the first base portion **428** of the first actuating member **426** defines a first cam groove **440** therein. The first cam groove **440** includes a first portion **442**, a second portion **444** spaced-apart from the first portion **442**, and a mid-portion **446** disposed between the first and second portions **442**, **444**. As shown, the mid-portion **446** of the first cam groove **440** is curved away from an axis BB' connecting the first and second portions **442**, **444** of the first cam groove **440**.

The second base portion **436** of the second actuating member **434** defines a second cam groove **448** therein. The second cam groove **448** includes a first portion **450**, a second portion **452** spaced-apart from the first portion **450**, and a mid-portion **454** disposed between the first and second portions **450**, **452**. As shown, the mid-portion **454** of the second cam groove **448** is curved away from an axis CC' connecting the first and second portions **450**, **452** of the second cam groove **448**. Additionally, the second cam groove **448** includes a third portion **456** extending from a distal end of the second portion **452**. The third portion **456** is arcuately disposed in relation to the second portion **452**. According to certain embodiments, the first, second, and mid-portions **442-446**, **450-454** of the first and second cam grooves **440**, **448** correspond with one another. However, it may be noted that other configurations of the first and second cam grooves **440**, **448** may be implemented by persons skilled in the art to realize the features of this invention.

The actuating mechanism **424** further includes a rotary actuator **458** communicably coupled with the processor **208** of the infotainment device **108**. The actuating mechanism **424** further includes a cam member **460** having a central portion **462** coupled to the rotary actuator **458**. The rotary actuator **458** operatively rotates the cam member **460** about its central portion **462** in response to control signal output by the processor **208**. The cam member **460** also has a primary pin **464** and a secondary pin **466**. The primary pin **464** is disposed at a first end **468** of the cam member **460**. The secondary pin **466** is disposed at a second end **470** of the cam member **460**. The primary pin **464** is disposed in engagement with the first and second cam grooves **440**, **448** from respective ones of the first and second actuating members **426**, **434**.

Additionally, as best shown in FIG. 6, the second link member **438** of the second actuating member **434** is slidably coupled to the first link member **430** of the first actuating member **426** with the help of a slot-pin arrangement in which each of the slots are denoted by numeral **414** while each of the sliding pins are denoted by numeral **416**. Moreover, as best shown in FIG. 6, the second link member **438** may be resiliently biased by a spring **472** coupled to the cover member **402**. The spring **472** facilitates a default position for each of the first and second actuating members **426**, **434** which in turn would cause the first and second sets of flow-guiding vanes **120a**, **120b** to be generally disposed in their respective default positions. The spring may assist secondary pin **466** in pulling the to pull the second link member **438** back to a centralized location. Further, when no user input is received at the user interface **206** for turning ON the HVAC system **110** or for effecting a particular flow-guiding position other than the vane closeout position, the default position of the first and second actuating members **426**, **434** may be effected by the spring **472** as the spring **472** generally biases the first and second actuating members **426**, **434** into their respective default positions, which

according to a particular embodiment of this disclosure, is the vane closeout position. Alternatively, it has also been contemplated that when a user input is received vis-à-vis the high-level menu option **202d** for turning OFF the HVAC system **110**, the spring **472** offers a bias force to each of the first and second actuating members **426**, **434** such that the first and second sets of flow-guiding vanes **120a**, **120b** are disposed in their respective vane closeout positions. It may be noted that the terms 'vane closeout position' disclosed herein may be regarded as a position in which the vanes **120** from the first and second sets of flow-guiding vanes **120a**, **120b** fully obstruct the flow of air from the air vent assembly **112** towards the interior **102** of the vehicle **100**.

According to certain embodiments, a rotation of the cam member **460** about an axis DD' of the central portion **462** causes an engagement of the primary pin **464** with one of the first, second, and mid-portions **442-446**, **450-454** of respective ones of the first and second cam grooves **440**, **448** for positioning the first and second sets of flow-guiding vanes **120a**, **120b** in one of: the vane closeout position, the rightward flow-guiding position, the onward flow-guiding position, and the leftward flow-guiding position. Explanation to each of these functions is made in the appended disclosure.

According to specific embodiments as shown in FIG. 8, when the primary pin **464** of the cam member **460** is operatively moved into engagement with the first portions **442**, **450** of respective ones of the first and second cam grooves **440**, **448** (refer to FIG. 4) and is located distally away from the mid-portions **446**, **454** of respective ones of the first and second cam grooves **440**, **448**, a corresponding movement of the first and second actuating members **426**, **434** causes respective ones of the first and second sets of flow-guiding vanes **120a**, **120b** to be positioned in their respective vane close-out positions.

According to other embodiments as shown in FIG. 9, when the primary pin **464** of the cam member **460** is operatively moved into engagement with the first portions **442**, **450** of respective ones of the first and second cam grooves **440**, **448** (refer to FIG. 4) so as to be located partway along a length of the first portions **442**, **450** of respective ones of the first and second cam grooves **440**, **448**, a corresponding movement of the first and second actuating members **426**, **434** causes respective ones of the first and second sets of flow-guiding vanes **120a**, **120b** to be positioned in their respective rightward flow-guiding positions. It may be noted that the flow-guiding vanes **120a**, **120b** as shown in FIG. 9 are pointing in a leftward and downward direction as the view of FIG. 9 is the bottom view of the air vent assembly **112**.

According to other embodiments as shown in FIG. 10, when the primary pin **464** of the cam member **460** is operatively moved into engagement with the first portions **442**, **450** of respective ones of the first and second cam grooves **440**, **448** (refer to FIG. 4) so as to be located proximal to the mid-portions **446**, **454** of respective ones of the first and second cam grooves **440**, **448**, a corresponding movement of the first and second actuating members **426**, **434** causes respective ones of the first and second sets of flow-guiding vanes **120a**, **120b** to be positioned in their respective onward flow-guiding positions.

According to other embodiments as shown in FIG. 11, when the primary pin **464** of the cam member **460** is operatively moved into engagement with the mid-portions **446**, **454** of respective ones of the first and second cam grooves **440**, **448** (refer to FIG. 4) and is located at a center of the mid-portions **446**, **454** of respective ones of the first

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and second cam grooves **440**, **448**, a corresponding movement of the first and second actuating members **426**, **434** causes respective ones of the first and second sets of flow-guiding vanes **120a**, **120b** to be positioned in their respective leftward flow-guiding positions. The flow-guiding vanes **120a**, **120b** as shown in FIG. **11** are pointing in a rightward and downward direction as the view of FIG. **9** is the bottom view of the air vent assembly **112**.

According to other embodiments as shown in FIG. **12**, when the primary pin **464** of the cam member **460** is operatively moved into engagement with the mid-portions **446**, **454** of respective ones of the first and second cam grooves **440**, **448** (refer to FIG. **4**) so as to be located proximal to the second portions **444**, **452** of respective ones of the first and second cam grooves **440**, **448**, a corresponding movement of the first and second actuating members **426**, **434** causes respective ones of the first and second sets of flow-guiding vanes **120a**, **120b** to be positioned in their respective onward flow-guiding positions.

According to certain embodiments, as shown in FIG. **13**, a rotation of the cam member **460** about the axis DD' of the central portion **462** causes a partially-split flow-guiding position, and if the rotation is continued further a fully-split flow-guiding position. When the primary pin **464** engages with the second portion **442** of the first cam groove **440** to be located distally away from the mid-portions **446**, **454** of the first and second cam grooves **440**, **448**. In this instance, upon further rotation of the cam member **460**, the first set of flow-guiding vanes **120b** rotate about their respective rotational axes AA' towards a leftward flow-guiding position. As the second cam groove **448** comprises a third portion **456**, the primary pin **464** of the rotating cam member **460** does not cause the second set of flow-guiding vanes **120a** to rotate. Rather, the secondary pin **466** of the cam member **460** engages with a retainer groove **474** defined on the second base portion **436**, causing the vanes **120** from the second set of flow-guiding vanes **120b** to rotate about their respective rotational axes AA' towards a rightward flow-guiding position.

For the purposes of this disclosure, the rightward and leftward flow-guiding positions of respective ones of the first and second sets of flow-guiding vanes **120a**, **120b** from the foregoing embodiments may be regarded as a split flow-guiding position in which angular positions of respective ones of the first and second flow-guiding vanes **120** may be dissimilar from angular positions of the first and second flow-guiding vanes **120** when both of the first and second flow-guiding vanes **120** are disposed in either of the leftward flow-guiding position or the rightward flow-guiding position. The terms 'partially-split flow-guiding position' disclosed herein is used to refer to a type of the split flow-guiding position in which the first and second sets of flow-guiding vanes **120a**, **120b** would be disposed at a first pre-defined angle with respect to one another. The first pre-defined angle disclosed herein may be, in an example, 45 degrees between the first and second sets of flow-guiding vanes **120a**, **120b**. In another example, the first pre-defined angle disclosed herein may be 60 degrees. The terms 'fully-split flow-guiding position' is however, another type of the split flow-guiding position in which the first and second sets of flow-guiding vanes **120a**, **120b** would be disposed at a second pre-defined angle with respect to one another, the second pre-defined angle being greater than the first pre-defined angle. In an example, if the first pre-defined angle is 45 degrees, then the second pre-defined angle may be implemented by way of a 90-degree angle between the first and second sets of flow-guiding vanes **120a**, **120b**. In

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another example, if the first pre-defined angle is 60 degrees, then the second pre-defined angle may be implemented by way of a 120-degree angle.

In another embodiment, the first and second pre-defined angles may result in the two sets of vanes being asymmetric with respect to one another. By selecting pre-determined lengths of the first and second cam grooves **440**, **448**, and in particular, a pre-determined length of the third portion **456** of the second cam groove **448** to achieve other values of the second pre-defined angle, such values however being greater in relation to the first pre-defined angle. For example, if the first pre-defined angle is 45 degrees, then the second pre-defined angle may be implemented by way of a 60-degree angle between the first and second sets of flow-guiding vanes **120a**, **120b**. In another example, the first pre-defined angle may be 45 degrees while the second pre-defined angle is 120 degrees.

According to certain other embodiments of this invention, the first and second pre-defined angles corresponding to the partially and fully split-flow-guiding positions of the first and second flow-guiding vanes **120** can be pre-defined to the user interface **206** by the user. For example, the user interface **206** of the infotainment device **108** can present to the user via other menu options (not shown)—a range of values for selecting each of the first and second pre-defined angles for subsequent implementation by the processor **208** for rotating the cam member **460** via the rotary actuator **458**.

FIG. **14A** illustrates cam-groove geometries for the pair of actuating members of the air vent assembly according to certain other embodiments of the invention. FIG. **14A** illustrates a cam-groove geometry for the first actuating member **426** that is linear and a cam groove geometry for the second actuating member **434** that is curved so as to define a "J" shaped portion **478** therein. FIG. **14B** illustrates a different set of cam-groove geometries for the pair of actuating members **426**, **434** of the air vent assembly **112** according to certain embodiments of the invention. FIG. **14B** illustrates a cam-groove geometry for the first actuating member **426** that is predominantly linear but includes a bend **480** and a cam-groove geometry for the second actuating member **434** that is curved in a "J" portion **482** and also includes a bend **484** in the linear portion **486** of the cam groove. The specific geometries of the cam grooves **490**, **492** defined on the first and second actuating members **426**, **434** may be different according to certain embodiments of the invention.

In both of the cam-groove geometries illustrated in FIGS. **14a** and **14b**, a portion of the cam-groove geometries is the same, and a portion of the cam-groove geometries is different. This allows for the single cam member **460** to cause the vanes **120** from both the first and second sets of flow-guiding vanes **120a**, **120b** to move in the same direction when the cam member **460** engages the portion of the cam grooves that are of the same geometry and also allows the cam member **460** to move the vanes **120** such that the vanes **120** associated with the first set of flow-guiding vanes **120a** are in a different direction than the vanes **120** associated with the second set of flow-guiding vanes **120b** when the primary pin **464** engages with the dissimilar portions from respective cam grooves of the first and second actuating members **426**, **434**. This is particularly important to produce the partial and full split flow-guiding positions disclosed herein.

The specific configurations of the first and second actuating members **426**, **434** described above to close the vanes **120** or move the vanes **120** from respective ones of the first and second sets of flow-guiding vanes **120a**, **120b** into the leftward, rightward, and split flow-guiding position is non-

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limiting of this disclosure. Rather, it will be acknowledged by persons skilled in the art that various other mechanisms may be used in place of the first and second actuating members 426, 434 disclosed herein without deviating from the spirit of the present disclosure.

The foregoing disclosure is not intended to limit the present disclosure to the precise forms or particular fields of use disclosed. As such, it is contemplated that various alternative embodiments and/or modifications to the present disclosure, whether explicitly described or implied herein, are possible in light of the disclosure. Having thus described embodiments of the present disclosure, a person of ordinary skill in the art will recognize that changes may be made in form and detail without departing from the scope of the present disclosure. Thus, the present disclosure is limited only by the claims.

In the foregoing specification, the disclosure has been described with reference to specific embodiments. However, as one skilled in the art will appreciate, various embodiments disclosed herein can be modified or otherwise implemented in various other ways without departing from the spirit and scope of the disclosure. Accordingly, this description is to be considered as illustrative and is for the purpose of teaching those skilled in the art the manner of making and using various embodiments of the disclosed air vent assembly. It is to be understood that the forms of disclosure herein shown and described are to be taken as representative embodiments. Equivalent elements, materials, processes or steps may be substituted for those representatively illustrated and described herein. Moreover, certain features of the disclosure may be utilized independently of the use of other features, all as would be apparent to one skilled in the art after having the benefit of this description of the disclosure. Expressions such as “including”, “comprising”, “incorporating”, “consisting of”, “have”, “is” used to describe and claim the present disclosure are intended to be construed in a non-exclusive manner, namely allowing for items, components or elements not explicitly described also to be present. Reference to the singular is also to be construed to relate to the plural.

Further, various embodiments disclosed herein are to be taken in the illustrative and explanatory sense, and should in no way be construed as limiting of the present disclosure. All joinder references (e.g., attached, affixed, coupled, connected, and the like) are only used to aid the reader's understanding of the present disclosure, and may not create limitations, particularly as to the position, orientation, or use of the systems and/or methods disclosed herein. Therefore, joinder references, if any, are to be construed broadly. Moreover, such joinder references do not necessarily infer that two elements are directly connected to each other.

Additionally, all numerical terms, such as, but not limited to, “first”, “second”, “third”, “primary”, “secondary”, “main” or any other ordinary and/or numerical terms, should also be taken only as identifiers, to assist the reader's understanding of the various elements, embodiments, variations and/or modifications of the present disclosure, and may not create any limitations, particularly as to the order, or preference, of any element, embodiment, variation and/or modification relative to, or over, another element, embodiment, variation and/or modification.

It will also be appreciated that one or more of the elements depicted in the drawings/figures can also be implemented in a more separated or integrated manner, or even removed or rendered as inoperable in certain cases, as is useful in accordance with a particular application. Additionally, any

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signal hatches in the drawings/figures should be considered only as exemplary, and not limiting, unless otherwise specifically specified.

The invention claimed is:

1. A system for controlling an air vent assembly within an HVAC system of a vehicle, the system comprising:

an actuating mechanism for independent control of a first set of flow guiding vanes and a second of flow guiding vanes, the actuating mechanism that comprises a single rotary actuator mechanically coupled to both the first set of flow guiding vanes and the second set of flow guiding vanes via a pair of actuating members comprising a first actuating member that comprises a first link member to engage with each vane of the first set of flow guiding vanes, and a second actuating member that comprises a second link member to engage with each vane of the second set of flow guiding vanes;

a display;

a processor in communication with the display;

a user interface in communication with the processor, the user interface configured to accept one or more inputs from a selection of menu control options;

wherein the processor is configured to cause display of a menu of control options; and

wherein at least one of the menu control options is an option to independently and simultaneously control, via the single rotary actuator, a positioning of the first set of flow guiding vanes and the second set of flow guiding vanes not directly connected to the first set of flow guiding vanes and arranged to be adjacent to each other in an elongated linear vent opening of the air vent assembly of the HVAC system.

2. The system of claim 1, wherein the menu control options include options for positioning respective ones of the first and second sets of flow guiding vanes in one of: a vane closeout position, a leftward flow-guiding position, an onward flow-guiding position, a rightward flow-guiding position, a partially-split flow-guiding position, and a fully-split flow-guiding position.

3. The system of claim 2, wherein in response to selection of one or more inputs via the user interface, the first set of flow guiding vanes and the second set of flow guiding vanes are independently moved to a position selected from the group consisting of: the vane closeout position, the leftward flow-guiding position, the onward flow-guiding position, the rightward flow-guiding position, the partially-split flow-guiding position, and the fully-split flow-guiding position.

4. The system of claim 3, wherein when the first set of flow guiding vanes and the second set of flow guiding vanes are independently moved to the partially-split flow-guiding position, the first and second sets of flow guiding vanes are disposed in a first pre-defined angular position relative to one another; and

when the first set of flow guiding vanes and the second set of flow guiding vanes are independently moved to the fully-split flow-guiding position, the first and second sets of flow guiding vanes are disposed in a second pre-defined angular position relative to one another, wherein the second pre-defined angular position is greater than the first pre-defined angular position.

5. The system of claim 4, wherein each of the first pre-defined angular position and the second pre-defined angular position is settable by a menu control option.

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6. The system of claim 1, wherein the air vent assembly comprises:

- a cover member having:
- a pair of elongated walls disposed in a mutually opposing relation to one another; and
- an inlet disposed on an aft side of the pair of mutually opposing elongated walls, the inlet configured to allow an entry of air into the cover member.

7. The system of claim 6, further comprising a vent extending from the inlet toward a fore side of the cover member, the vent configured to allow air to exit the cover member, the vent comprising:

- a first vent portion having the first set of flow guiding vanes disposed therein, each vane from the first set of flow guiding vanes pivotally coupled to the pair of mutually opposing elongated walls of the cover member; and
- a second vent portion located adjacent to the first vent portion, the second vent portion having the second set of flow guiding vanes disposed therein, each vane from the second set of flow guiding vanes pivotally coupled to the pair of mutually opposing elongated walls of the cover member.

8. The system of claim 1, wherein the one or more inputs from the selection of menu control options includes a voice command.

9. The system of claim 1, wherein the user interface comprises a touch screen.

10. The system of claim 1, wherein the at least one of the menu control options includes the option to independently and simultaneously control the positioning of the first set of flow guiding vanes and the second set of flow guiding vanes in the air vent assembly.

11. The system of claim 1, wherein the processor controls the rotary actuator based on a selected option.

12. The system of claim 1, wherein the actuating mechanism further comprises:

- a first actuating member comprising a first link member configured to engage with each vane of the first set of flow guiding vanes; and
- a second actuating member comprising a second link member configured to engage with each of the vanes from the second set of flow guiding vanes.

13. The system of claim 1, wherein the user interface is disposed at a mid-portion of a vehicle dash console.

14. The system of claim 1, wherein the single rotary actuator is located adjacent to a first non-rotary actuating member, and the first non-rotary actuating member is further arranged to be adjacent to a second non-rotary actuating member.

15. The system of claim 1, additionally comprising a cam member having a central portion adjacently coupled to the single rotary actuator.

16. A vehicle comprising:

- a system for controlling an air vent assembly within an HVAC system of the vehicle, the system comprising:
- an actuating mechanism for independent control of a first set of flow guiding vanes and a second of flow guiding vanes, the actuating mechanism that comprises a single rotary actuator mechanically coupled to both the first set of flow guiding vanes and the second set of flow guiding vanes via a pair of actuating members comprising a first actuating member that comprises a first link member to engage with each vane of the first set of flow guiding vanes, and a second actuating member that comprises a second link member to engage with each vane of the second set of flow guiding vanes;

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a processor;

a user interface in communication with the processor, the user interface configured to accept one or more inputs associated with selection of menu control options;

wherein the processor is configured to output a menu of options; and

wherein at least one of the menu control options is an option to independently and simultaneously control, via the single rotary actuator, a positioning of the first set of flow guiding vanes and the second set of flow guiding vanes not directly connected to the first set of flow guiding vanes and arranged to be adjacent to each other in an elongated linear vent opening of the air vent assembly of the HVAC system.

17. The vehicle of claim 16, wherein in response to selection of one or more inputs associated with the menu control options, the first set of flow guiding vanes and the second set of flow guiding vanes are independently moved to a position selected from the group consisting of: a vane closeout position, a leftward flow-guiding position, an onward flow-guiding position, a rightward flow-guiding position, a partially-split flow-guiding position, and a fully-split flow-guiding position.

18. The vehicle of claim 16, wherein the user interface comprises a touch screen.

19. The vehicle of claim 16, wherein the processor is operatively coupled to the rotary actuator.

20. The vehicle of claim 16, wherein the processor controls the rotary actuator based on a selected option.

21. The vehicle of claim 16, wherein the air vent assembly comprises:

- a cover member comprising an inlet;
- a vent extending from the inlet, the vent comprising the first set of flow guiding vanes and the second set of flow guiding vanes; and
- an actuating mechanism, configured to receive a control signal from the processor, operably coupled to each of the first set of flow guiding vanes and the second set of flow guiding vanes.

22. The vehicle of claim 21, wherein the actuating mechanism comprises:

- a first actuating member comprising a first link member configured to engage with each vane of the first set of flow guiding vanes, wherein the rotary actuator is located adjacent to the first actuating member; and
- a second actuating member comprising a second link member configured to engage with each of the vanes from the second set of flow guiding vanes, such that the first set of flow guiding vanes and the second set of flow guiding vanes are capable of moving independently of one another, wherein the first actuating member is further arranged to be adjacent to the second actuating member.

23. A vehicle comprising:

- a system for controlling an air vent assembly within an HVAC system of the vehicle, the system comprising:
- a processor;

a user interface in communication with the processor, the user interface configured to accept one or more inputs associated with selection of menu control options;

wherein the processor is configured to output a menu of options;

wherein at least one of the menu options is an option to independently and simultaneously control a positioning of a first set of flow guiding vanes and a separate second set of flow guiding vanes not directly connected to the first set of flow guiding vanes and arranged to be

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adjacent to each other in a horizontal direction in an elongated linear vent opening of the air vent assembly of the HVAC system;

an actuating mechanism, configured to receive a control signal from the processor, for independent control of the first set of flow guiding vanes and the second set of flow guiding vanes;

wherein the actuating mechanism comprises:

a single rotary actuator configured to receive the control signal from the processor;

a pair of actuating members coupled with the single rotary actuator, the pair of actuating members including a first actuating member and a second actuating member;

the first actuating member coupled with the single rotary actuator and comprising a first link member configured to engage with each vane of the first set of flow guiding vanes, wherein the single rotary actuator is located adjacent to the first actuating member; and

a second actuating member coupled with the single rotary actuator and comprising a second link member configured to engage with each of the vanes from the second set of flow guiding vanes, such that the first set of flow guiding vanes and the second set of flow guiding vanes are capable of moving independently of one another, wherein the first actuating member is further arranged to be adjacent to the second actuating member.

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